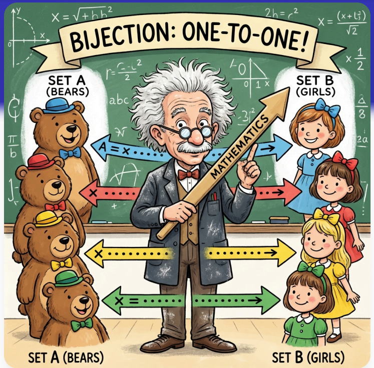
[illegible] [illegible] [illegible]

[illegible] [illegible] [illegible] [illegible] :[illegible] [illegible]

[illegible] [illegible] .[illegible] :[illegible]

[illegible]

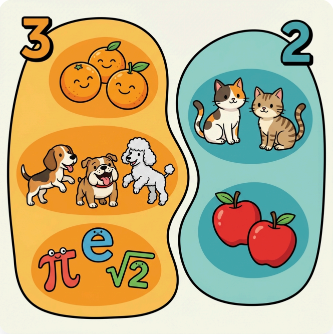

BIJECTION: ONE-TO-ONE!
SET A (BEARS)
SET B (GIRLS)
MATHEMATICS
SET A (BEARS)
SET B (GIRLS)

[illegible]: [illegible] A, B [illegible]. [illegible] $A \sim B$ [illegible] $f : A \to B$.

[illegible]: [illegible] $\sim$ [illegible].

[illegible]:

1. **[illegible]:** [illegible] A:
   - [illegible] $Id_A(x) = x$.
   - [illegible].
   - [illegible] $A \sim A$.
2. **[illegible]:** [illegible] $A \sim B$.
   - [illegible] $f : A \to B$ [illegible].
   - [illegible] $f^{-1} : B \to A$.
   - [illegible] f^{-1} [illegible].
   - [illegible] $B \sim A$.
3. **[illegible]:** [illegible] $A \sim B$ [illegible]-$B \sim C$.
   - [illegible] $f : A \to B$ [illegible]-$g : B \to C$.
   - [illegible] $h = g \circ f : A \to C$.
   - [illegible].
   - [illegible] $A \sim C$.



[illegible]
[illegible]:

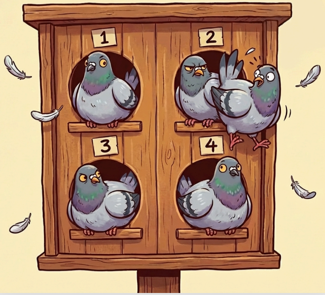
[illegible]

- [illegible] $\mathbb{N}^{<n}$: [illegible] n [illegible]. [illegible] $\mathbb{N}^{<n} = \{0, 1, \ldots, n-1\}$.
  - [illegible] $\mathbb{N}^{<0} = \emptyset$.
- [illegible]: [illegible] $S \subseteq \mathbb{N}$. [illegible] S [illegible] (bounded) [illegible] $n \in \mathbb{N}$ [illegible] $S \subseteq \mathbb{N}^{<n}$.
- [illegible]:
  - [illegible] 12, $\{1, 2, 3, 4, 6, 12\}$, [illegible] $\mathbb{N}$ [illegible] ($\mathbb{N}^{<13}$ [illegible]).
  - [illegible] $\emptyset$ [illegible] ($\mathbb{N}^{<0}$ [illegible]).
  - [illegible] $\{0, 1, \ldots, 9\}$ [illegible] ($\mathbb{N}^{<10}$ [illegible]).
  - [illegible] $\{0, 2, 4, 6, \ldots\}$ [illegible].

[illegible]

- [illegible]: [illegible] n [illegible] m-[illegible], [illegible] $n > m$, [illegible].
- [illegible]: [illegible] $n > m$ [illegible]. [illegible] $f : \mathbb{N}^{<n} \to \mathbb{N}^{<m}$.
- [illegible]: [illegible] $n \neq m$, [illegible] (bijection) [illegible] $\mathbb{N}^{<n}$ [illegible]-$\mathbb{N}^{<m}$.
- [illegible]:
  - [illegible] $n \neq m$. [illegible] $n > m$.
  - [illegible], [illegible] $\mathbb{N}^{<n}$-[illegible] $\mathbb{N}^{<m}$-[illegible].
  - [illegible].
  - [illegible].
- [illegible].
  [illegible] – [illegible]!

[illegible] 4-[illegible] 5
[illegible]



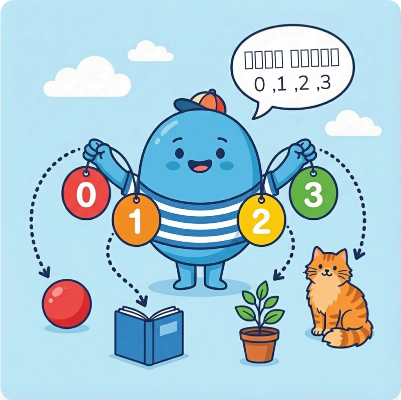
[illegible] [illegible]

[illegible]: [illegible] A [illegible] [illegible] [illegible]. A [illegible] [illegible] [illegible] $n \in \mathbb{N}$ [illegible] [illegible] [illegible]"[illegible] (bijection) [illegible]-A [illegible]-$\mathbb{N}^{<n}$.

- [illegible] [illegible], [illegible] $|A| = n$ [illegible] [illegible]-"A [illegible] [illegible] n" [illegible] "A [illegible] n [illegible]".
- [illegible] [illegible] [illegible] [illegible] .

[illegible]:

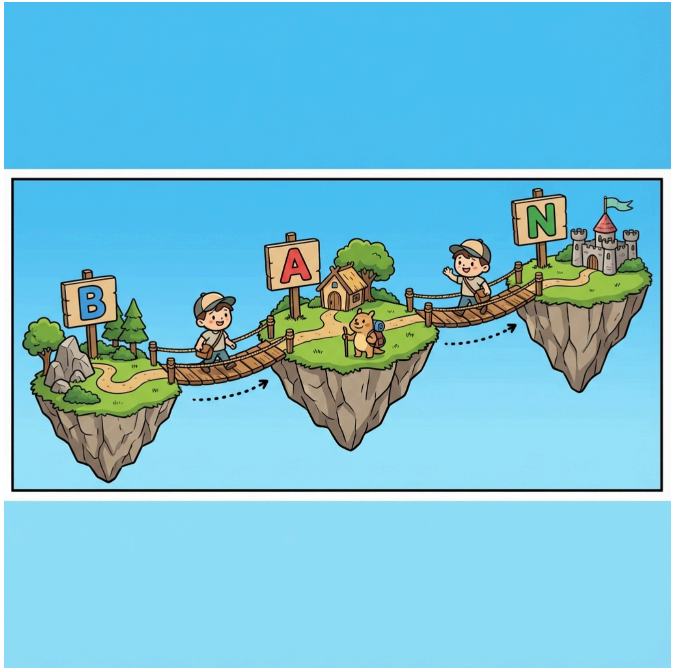
- [illegible] $\{a, b, c\}$ [illegible] [illegible] [illegible] 3 [illegible].
- [illegible] $\mathbb{N}$ [illegible] [illegible].
- [illegible] [illegible] $\emptyset$ [illegible] [illegible] [illegible] 0 [illegible].



[illegible]

[illegible]: [illegible] A [illegible] [illegible]-$A \sim B$, [illegible] B [illegible] [illegible]-$|B| = |A|$.

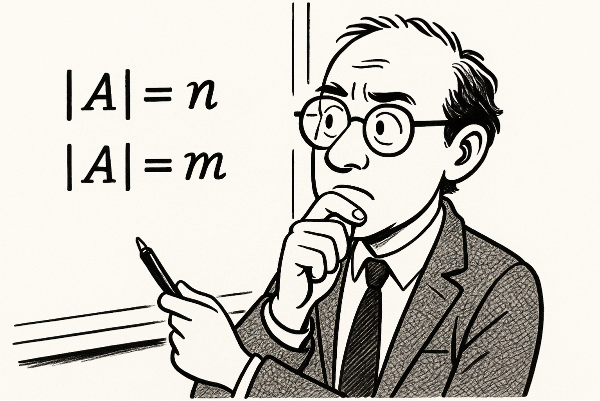
[illegible]:



- [illegible] [illegible]-$|A| = n$. [illegible] [illegible]-B [illegible] [illegible]-$|B| = n$.
- [illegible]-$|A| = n$, [illegible] [illegible] [illegible] [illegible] $f\colon A \to \mathbb{N}^{<n}$.
- [illegible]-$A \sim B$, [illegible] [illegible] [illegible] [illegible] $g\colon A \to B$.
- [illegible] [illegible] $f \circ g^{-1}\colon B \to \mathbb{N}^{<n}$.
- [illegible]-g [illegible], [illegible] g^{-1} [illegible].
- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].
- [illegible] $f \circ g^{-1}$ [illegible] [illegible].
- [illegible] [illegible]-$B \sim \mathbb{N}^{<n}$, [illegible] B [illegible]-$|B| = n$.

[illegible] [illegible]

- [illegible]:
  - $f: A \to \mathbb{N}^{<n}$
  - $g: A \to \mathbb{N}^{<m}$
- [illegible] $g \circ f^{-1}$ [illegible] [illegible]-$\mathbb{N}^{<n}$ [illegible]-$\mathbb{N}^{<m}$.
- [illegible], [illegible] $n = m$.
- [illegible]: [illegible].



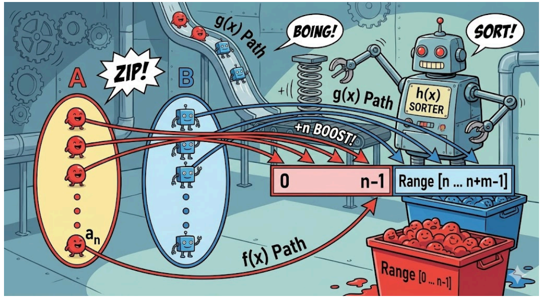
[illegible]

[illegible]: [illegible] A, B [illegible], [illegible]:

$$|A \cup B| = |A| + |B|$$

[illegible]:

- [illegible] $|A| = n$ [illegible]-$|B| = m$.
- [illegible] $f\colon A \to \mathbb{N}^{<n}$ [illegible]-$g\colon B \to \mathbb{N}^{<m}$.
- [illegible] $h\colon A \cup B \to \mathbb{N}^{<n+m}$ [illegible]:


A
ZIP!
B
g(x) Path
BOING!
g(x) Path
SORT!
h(x) SORTER
+n BOOST!
0
n-1
Range [n … n+m-1]
a_n
f(x) Path
Range [0 … n-1]

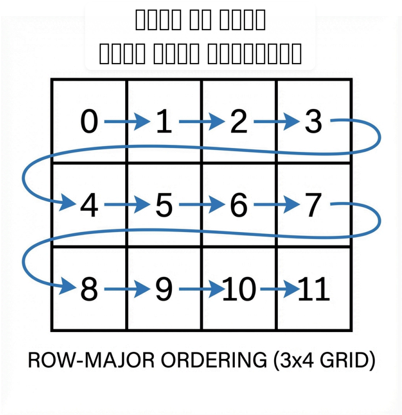
$$h(x) = \begin{cases} f(x) & x \in A \\ g(x) + n & x \in B \end{cases}$$

- [illegible]-A [illegible]-B [illegible], [illegible] h [illegible].
- [illegible]-h [illegible], [illegible] $|A \cup B| = n + m$.

□□□□□□□□ □□□□□□□ □□□□□

□□□□: □□ A, B □□□□□□□ □□□□□□□, □□:

$$|A \times B| = |A| \cdot |B|$$



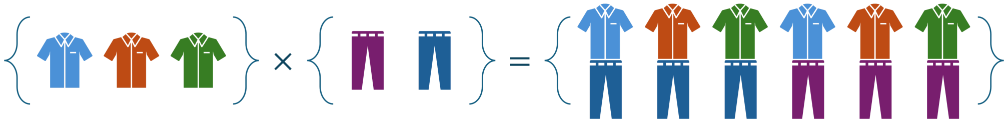
ROW-MAJOR ORDERING (3x4 GRID)

□□□□□□:

- □□□□ $|A| = n$ □-$|B| = m$.
- □□□□□□□ □□□□□□□□□ □"□□ □□□ $f: A \to \mathbb{N}^{<n}$ □-$g: B \to \mathbb{N}^{<m}$.
- □□□□□□□ □□□□□□□□ $h: A \times B \to \mathbb{N}^{<nm}$ □□□□□□ □□□□:

$$h(a,b) = f(a) \cdot m + g(b)$$

- □□□□ □□□□□□ □□□□□□□ □□□□□□□□ □□ (a,b) □□□□□□ □□□□□□ $n \times m$ (□□□□□□□□□□ □□□□□ □□□□ □□□□□).
- □□□□□□□□ □□□□□□ □□ □□□□□ □□ □□□□□□ □□□□□□□□ □□□□ □□□□□□□ (□□□□□ □□□□□□□ □□ □□□□□□), h □□□□ □"□□ □□□□.
- □□□ $|A \times B| = n \cdot m$.



[illegible]

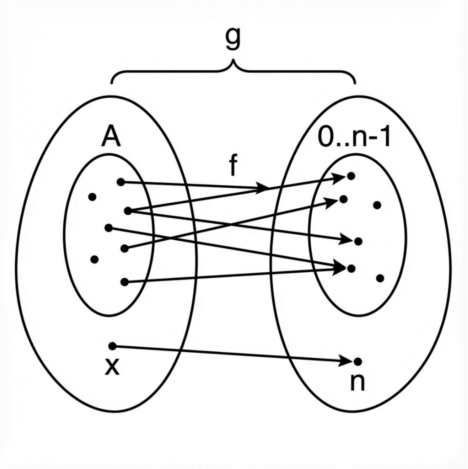
[illegible]: [illegible] A [illegible] n, [illegible] $x \notin A$. [illegible] $A \cup \{x\}$ [illegible] $n+1$.

[illegible]:

- [illegible]-$|A| = n$, [illegible] $f : A \to \mathbb{N}^{<n}$.
- [illegible] $g : A \cup \{x\} \to \mathbb{N}^{<n+1}$ [illegible]:

$$g(z) = \begin{cases} f(z) & z \in A \\ n & z = x \end{cases}$$

- [illegible] $x \notin A$.
- [illegible]: $g(z_1) = g(z_2)$ [illegible]:
  - [illegible] n, [illegible] x.
  - [illegible]-$\mathbb{N}^{<n}$, [illegible]-A [illegible]-f.
- [illegible]:
  - n [illegible]-x.
  - [illegible]-$\mathbb{N}^{<n}$ [illegible]-A ([illegible] f [illegible]).



[illegible] $n \in \mathbb{N}$?

[illegible] [illegible] [illegible] [illegible]

[illegible]: [illegible] $|A| = n+1$ [illegible]-$a \in A$, [illegible] $|A \setminus \{a\}| = n$.

[illegible] [illegible] [illegible] ([illegible] [illegible]):

1. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $f|_X\colon X \to f[X]$.
2. [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

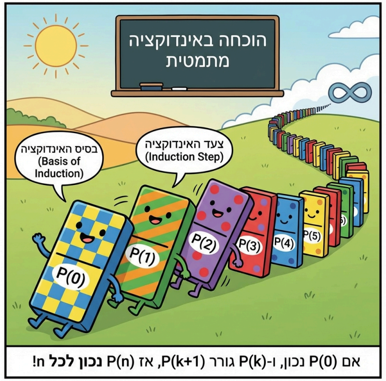
[illegible] – [illegible], [illegible] [illegible]:

- [illegible] $f\colon A \to \mathbb{N}^{<n+1}$ [illegible].
- [illegible] 1: $f(a) = n$.
- [illegible] $f[A \setminus \{a\}] = \mathbb{N}^{<n+1} \setminus \{n\} = \mathbb{N}^{<n}$.
- [illegible] [illegible] 1, $f|_{A\setminus\{a\}}\colon A \setminus \{a\} \to \mathbb{N}^{<n}$ [illegible].

- [illegible] 2: $f(a) = k \neq n$.
- [illegible] $c = f^{-1}(n)$. [illegible] $c \neq a$.
- [illegible] $g = f|_{A\setminus\{a,c\}} \cup \{\langle c, k\rangle\}$.
- [illegible] [illegible] 1, $f|_{A\setminus\{a,c\}}$ [illegible] [illegible] $\mathbb{N}^{<n} \setminus \{k\}$.
- $\{\langle c, k\rangle\}$ [illegible] [illegible]-$\{c\}$ [illegible]-$\{k\}$.
- [illegible] [illegible] 2, $g\colon A \setminus \{a\} \to \mathbb{N}^{<n}$ [illegible].

[illegible]: [illegible] [illegible] [illegible] [illegible] ([illegible] – [illegible] [illegible] k, [illegible] [illegible]) [illegible]: [illegible] $n \leq k$ [illegible] [illegible] [illegible] n, [illegible] $n-k$ [illegible].

[illegible]


הוכחה באינדוקציה מתמטית
בסיס האינדוקציה (Basis of Induction)
צעד האינדוקציה (Induction Step)
P(0)
P(1)
P(2)
P(3)
P(4)
P(5)
אם P(0) נכון, ו-P(k) גורר P(k+1), אז P(n) נכון לכל n!

- [illegible] $P(n)$:
  - $P(0)$ [illegible]
  - [illegible] n, [illegible] $P(n)$ [illegible] $P(n+1)$ [illegible]

- [illegible], [illegible], [illegible] $n = 0$
  - $P(0) \implies P(1)$
  - [illegible]-$P(0)$ [illegible], [illegible] $P(1)$ [illegible].
- [illegible], [illegible], [illegible] $n = 1$
  - $P(1) \implies P(2)$
  - [illegible]-$P(1)$ [illegible], [illegible] $P(2)$ [illegible].
- [illegible], [illegible], [illegible] $n = 2$
  - $P(2) \implies P(3)$
  - [illegible]-$P(2)$ [illegible], [illegible] $P(3)$ [illegible].
- [illegible].
- [illegible]: [illegible] $\forall n \in \mathbb{N}(P(n))$.

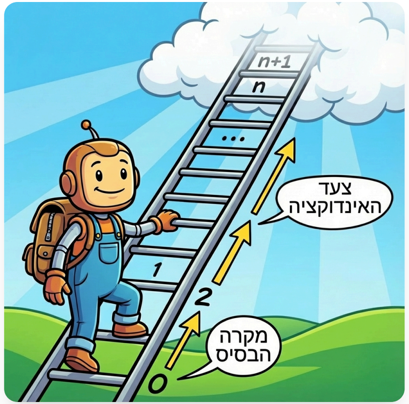
[illegible]?

[illegible] [illegible] ([illegible] [illegible])

[illegible] $A \subseteq \mathbb{N}$ [illegible] [illegible] [illegible] [illegible] [illegible]:

1. $0 \in A$
2. [illegible] $a \in A$ [illegible] $a + 1 \in A$

[illegible]: $A = \mathbb{N}$.


n+1
n
...
צעד האינדוקציה
1
2
מקרה הבסיס
0

[illegible]: [illegible] $n \in \mathbb{N}$ [illegible] $\sum_{i=0}^{n} i = \frac{n(n+1)}{2}$.

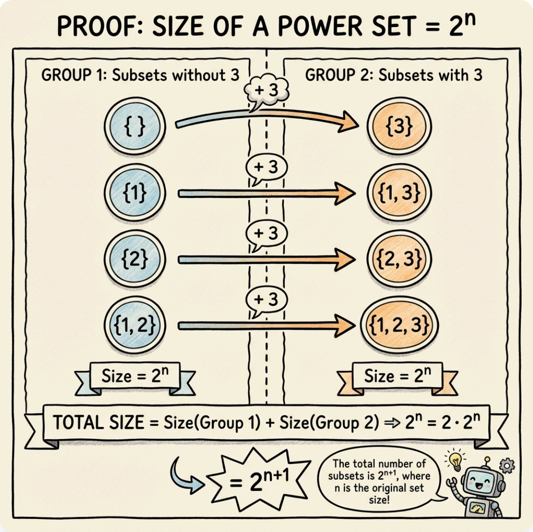
[illegible]:

- [illegible] $A = \{n \in \mathbb{N} \mid \sum_{i=0}^{n} i = \frac{n(n+1)}{2}\}$.
- [illegible]: [illegible] $n = 0$, [illegible] [illegible] 0 [illegible] [illegible] $\frac{0(1)}{2} = 0$. [illegible] $0 \in A$.
- [illegible]: [illegible] $n \in A$. [illegible] $n + 1 \in A$:

$$\sum_{i=0}^{n+1} i = \left(\sum_{i=0}^{n} i\right) + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$$

- [illegible] $n + 1 \in A$. [illegible] [illegible] [illegible], $A = \mathbb{N}$.

[illegible]


PROOF: SIZE OF A POWER SET = 2ⁿ
GROUP 1: Subsets without 3
GROUP 2: Subsets with 3
{ }
{3}
+3
{1}
{1, 3}
+3
{2}
{2, 3}
+3
{1, 2}
{1, 2, 3}
+3
Size = 2ⁿ
Size = 2ⁿ
TOTAL SIZE = Size(Group 1) + Size(Group 2) ⇒ 2ⁿ = 2 · 2ⁿ
= 2ⁿ⁺¹
The total number of subsets is 2ⁿ⁺¹, where n is the original set size!

[illegible]: [illegible] $|A| = n$, [illegible] $|\mathcal{P}(A)| = 2^n$.

[illegible] n:

[illegible] $(n = 0)$: $A = \emptyset$, $\mathcal{P}(A) = \{\emptyset\}$. [illegible] $h(\emptyset) = 0$. $h\colon \mathcal{P}(\emptyset) \to \mathbb{N}^{<1}$ [illegible].

[illegible] $(n \to n+1)$:

- [illegible] A [illegible] n, [illegible] $f\colon \mathcal{P}(A) \to \mathbb{N}^{<2^n}$.
- [illegible] B [illegible] $n+1$. [illegible] $b \in B$ [illegible] $A = B \setminus \{b\}$ ([illegible] n, [illegible]).
- [illegible] $g\colon \mathcal{P}(B) \to \mathbb{N}^{<2^{n+1}}$ [illegible]:

$$g(X) = \begin{cases} f(X) & b \notin X \\ f(X \setminus \{b\}) + 2^n & b \in X \end{cases}$$

- [illegible] g-[illegible] ([illegible] b [illegible], [illegible] b [illegible]).
- [illegible] $|\mathcal{P}(B)| = 2^{n+1}$.

$\mathcal{P}(A) \sim \{0,1\}^A$

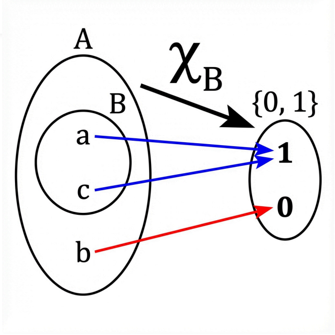
[illegible]: [illegible] A ([illegible]) [illegible] $\mathcal{P}(A) \sim \{0,1\}^A$.

[illegible]:

- [illegible] $B \subseteq A$, [illegible] $\chi_B\colon A \to \{0,1\}$:

$$\chi_B(x) = \begin{cases} 1 & x \in B \\ 0 & x \notin B \end{cases}$$

B [illegible] χ_B [illegible]



- [illegible] $H\colon \mathcal{P}(A) \to \{0,1\}^A$ [illegible]"[illegible] $H(B) = \chi_B$.
- [illegible]"[illegible]: [illegible] $B \neq C$, [illegible] x [illegible], [illegible] $\chi_B(x) \neq \chi_C(x)$.
- [illegible]: [illegible] $h\colon A \to \{0,1\}$ [illegible] $h = \chi_B$ [illegible] $B = \{x \in A \mid h(x) = 1\}$.
- [illegible]: [illegible] $|A| = n$, [illegible] $|\{0,1\}^A| = |\mathcal{P}(A)| = 2^n$.

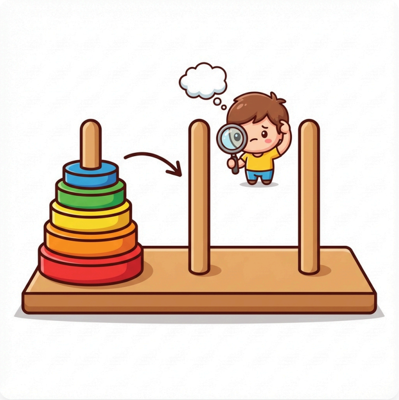
[illegible]: [illegible]

[illegible]: [illegible] 3 [illegible] n-[illegible]. [illegible], [illegible].

[illegible]: [illegible] n [illegible] $2^n - 1$.

[illegible]:

- **[illegible]:** [illegible] $n = 1$, [illegible]. $2^1 - 1 = 1$. [illegible].
- **[illegible]:** [illegible] n, [illegible] $n + 1$:
  - [illegible], [illegible] n [illegible].
  - [illegible], [illegible] $2^n - 1$ [illegible].
  - [illegible] (1 [illegible]).
  - [illegible], [illegible] n [illegible]. [illegible], [illegible], $2^n - 1$ [illegible].
  - [illegible]: $(2^n - 1) + 1 + (2^n - 1) = 2 \cdot 2^n - 1 = 2^{n+1} - 1$.



[illegible]

[illegible]: "[illegible], [illegible]".

1. [illegible]: [illegible], [illegible].
2. [illegible]:
   - [illegible] $k-1$ [illegible], [illegible] k [illegible].
   - [illegible] k [illegible]. [illegible].
   - [illegible] $k-1$ [illegible], [illegible].
   - [illegible].
   - [illegible] $k-1$ [illegible], [illegible].
   - [illegible] k-[illegible] [illegible].

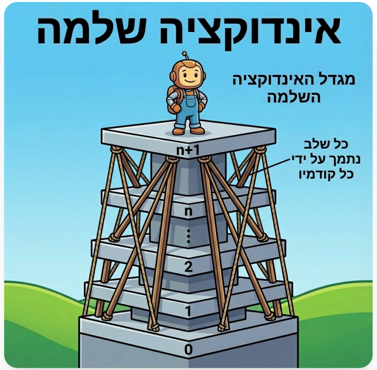
[illegible] ([illegible])

[illegible] $A \subseteq \mathbb{N}$ [illegible]:

[illegible] $n \in \mathbb{N}$,

$$\mathbb{N}^{<n} \subseteq A \implies \mathbb{N}^{<n+1} \subseteq A$$

[illegible]: $A = \mathbb{N}$.

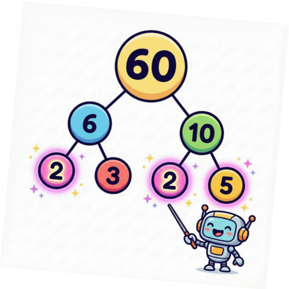

אינדוקציה שלמה
מגדל האינדוקציה השלמה
כל שלב נתמך על ידי כל קודמיו
n+1
n
⋮
2
1
0

[illegible]: [illegible].

[illegible]: [illegible] $n \geq 2$ [illegible].

[illegible]:

- [illegible] $A = \{n \in \mathbb{N} \mid n < 2 \vee$ [illegible]$\}$.
- [illegible] $n \in \mathbb{N}$. [illegible]-$\mathbb{N}^{<n} \subseteq A$. [illegible]-$n \in A$:
  - [illegible] $n < 2$: $n \in A$ [illegible].
  - [illegible] n [illegible]: n [illegible] $n \in A$.
  - [illegible] n [illegible]: $n = a \cdot b$ [illegible] $2 \leq a, b < n$.
    - [illegible]-$a, b < n$, [illegible]-$a, b \in \mathbb{N}^{<n} \subseteq A$.
    - [illegible] a, b [illegible] $\Longleftarrow n$ [illegible].
- [illegible] $A = \mathbb{N}$.

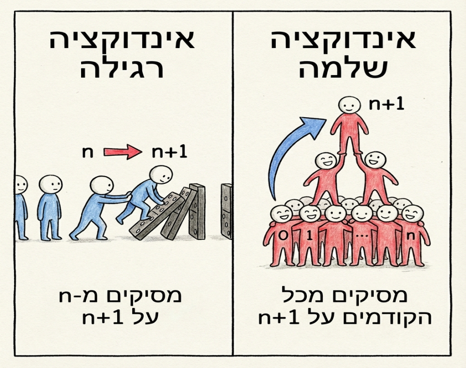


[illegible]?

$$\forall A\subseteq\mathbb{N}\left([0\in A\wedge\forall n\in\mathbb{N}(n\in A\to n+1\in A)]\to A=\mathbb{N}\right)$$

$$\Updownarrow\quad ?$$

$$\forall A\subseteq\mathbb{N}\left(\forall n\in\mathbb{N}(\mathbb{N}^{<n}\subseteq A\to\mathbb{N}^{<n+1}\subseteq A)\to A=\mathbb{N}\right)$$


אינדוקציה רגילה
n
n+1
מסיקים מ-n על n+1
אינדוקציה שלמה
n+1
0
1
...
n
מסיקים מכל הקודמים על n+1

[illegible]

[illegible] ⇐ [illegible]:

[illegible].

[illegible] A [illegible].

- [illegible] $B = \{n \in \mathbb{N} \mid \mathbb{N}^{<n} \subseteq A\}$.
- [illegible]: $0 \in B$ [illegible] $\mathbb{N}^{<0} = \emptyset \subseteq A$.
- [illegible]: [illegible]
  - $n \in B$ [illegible] $\mathbb{N}^{<n} \subseteq A$.
  - [illegible] A, [illegible] $\mathbb{N}^{<n+1} \subseteq A$.
  - [illegible] $n + 1 \in B$-[illegible].
- [illegible], $B = \mathbb{N}$.
- [illegible] n, [illegible] $n + 1 \in B$-[illegible], [illegible] $\mathbb{N}^{<n+1} \subseteq A$,
  - [illegible] $n \in A$.
- [illegible] $A = \mathbb{N}$.

[illegible] ⇐ [illegible]:

[illegible].

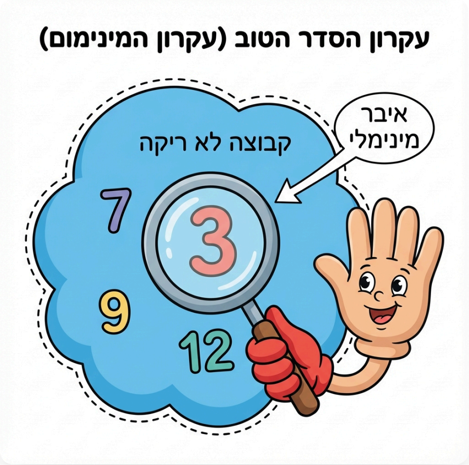
[illegible] A [illegible].

- [illegible] $n \in \mathbb{N}$. [illegible] $\mathbb{N}^{<n} \subseteq A$-[illegible]. [illegible] $n \in A$-[illegible]:
  - [illegible] $n = 0$: [illegible] $0 \in A$.
  - [illegible] $n > 0$: [illegible] $n - 1 \in \mathbb{N}^{<n}$, [illegible] $n - 1 \in A$.
  - [illegible], $n - 1 \in A \implies n \in A$.
- [illegible] $\mathbb{N}^{<n} \subseteq A \implies n \in A$-[illegible],
  - [illegible] A [illegible].
- [illegible] $A = \mathbb{N}$.

[illegible] [illegible] [illegible] ([illegible] [illegible])

[illegible] [illegible] [illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $\mathbb{N}$ [illegible] [illegible] [illegible] ([illegible]).

[illegible]: [illegible] $\emptyset \neq A \subseteq \mathbb{N}$, [illegible] [illegible] $a \in A$ [illegible] [illegible]-$a \leq x$ [illegible] $x \in A$.

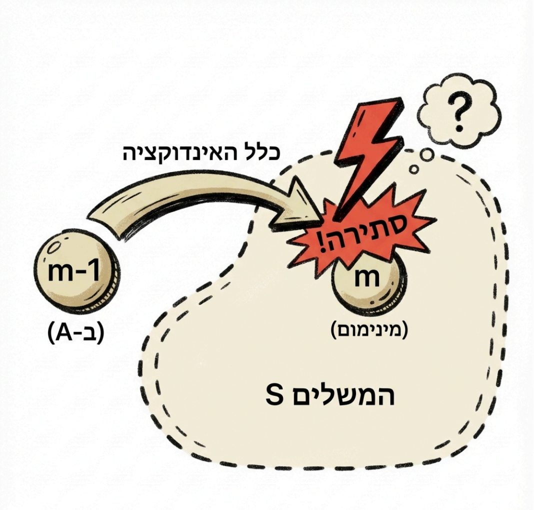

עקרון הסדר הטוב (עקרון המינימום)
קבוצה לא ריקה
איבר מינימלי
7
3
9
12

[illegible] [illegible] [illegible] ⇐ [illegible] [illegible]

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible]:

- [illegible] [illegible] [illegible] [illegible] [illegible].
- [illegible] $A \subseteq \mathbb{N}$ [illegible] $0 \in A$ [illegible] $\forall n(n \in A \to n+1 \in A)$.
- [illegible] $S = \mathbb{N} \setminus A$. [illegible] [illegible] [illegible]-$S \neq \emptyset$.
- [illegible] [illegible] [illegible] [illegible], [illegible] [illegible]-S [illegible] $m = \min(S)$.
- [illegible] $m = 0$: [illegible] [illegible] [illegible]-$0 \in A$.
- [illegible] $m > 0$: [illegible] [illegible]-m [illegible] [illegible]-S, [illegible] [illegible]-$m-1 \notin S$, [illegible] $m-1 \in A$.
- [illegible] [illegible] [illegible], $m-1 \in A \implies m \in A$, [illegible] $m \notin S$. [illegible]!
- [illegible] $S = \emptyset$ [illegible]-$A = \mathbb{N}$.


?
כלל האינדוקציה
סתירה!
m-1
(ב-A)
m
(מינימום)
המשלים S

[illegible] [illegible] $\Leftarrow$ [illegible] [illegible] [illegible]

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible]:

- [illegible] [illegible] [illegible]. [illegible] $S \subseteq \mathbb{N}$ [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible]-$S = \emptyset$.
- [illegible] $A = \mathbb{N} \setminus S$. [illegible] [illegible]-$A = \mathbb{N}$ [illegible] [illegible].
- [illegible] [illegible] $\mathbb{N}^{<n} \subseteq A$ ([illegible] [illegible] [illegible] [illegible] [illegible]-n [illegible] [illegible]-S).
- [illegible] $n \in S$, [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] S ([illegible] [illegible] [illegible]-S [illegible] [illegible] [illegible]).
- [illegible] [illegible]-S [illegible] [illegible], [illegible] $n \notin S$, [illegible] $n \in A$.
- [illegible] $\mathbb{N}^{<n} \subseteq A \implies \mathbb{N}^{<n+1} \subseteq A$. [illegible] [illegible] [illegible], $A = \mathbb{N}$.
- [illegible] $S = \emptyset$.

[illegible] [illegible] [illegible]-$\mathbb{N}$ [illegible] [illegible]

[illegible]: [illegible] [illegible] [illegible] $S \subseteq \mathbb{N}$ [illegible] [illegible].

[illegible]: [illegible] [illegible] [illegible] n [illegible] [illegible] $P(n)$: [illegible] $S \subseteq \mathbb{N}^{<n}$ [illegible] [illegible].

- [illegible] ($n = 0$): [illegible] $S \subseteq \mathbb{N}^{<0} = \emptyset$ [illegible] $S = \emptyset$, [illegible] [illegible] ([illegible] 0).
- [illegible] ($n \rightarrow n + 1$): [illegible] $P(n)$ [illegible] $S \subseteq \mathbb{N}^{<n+1}$.
  - [illegible] $S \setminus \{n\} \subseteq \mathbb{N}^{<n}$, [illegible] [illegible] [illegible] [illegible] $S \setminus \{n\}$ [illegible].
  - [illegible] $n \notin S$: $S = S \setminus \{n\}$, [illegible] S [illegible].
  - [illegible] $n \in S$: $S = (S \setminus \{n\}) \cup \{n\}$ [illegible] $n \notin S \setminus \{n\}$.

  [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], S [illegible] ([illegible] $|S \setminus \{n\}| + 1$).
- [illegible] $P(n)$ [illegible] [illegible] $n \in \mathbb{N}$.
- [illegible], [illegible] S [illegible], [illegible] n [illegible] [illegible]-$S \subseteq \mathbb{N}^{<n}$, [illegible] $P(n)$ [illegible] S [illegible].

[illegible]-[illegible] [illegible] [illegible] $\mathbb{N}$ [illegible] [illegible]

[illegible]: [illegible] $S \subseteq \mathbb{N}$ [illegible] [illegible], [illegible] S [illegible].

[illegible]: [illegible] [illegible] [illegible] n [illegible] [illegible] $P(n)$: [illegible] $S \subseteq \mathbb{N}$ [illegible] $|S| = n$ [illegible] [illegible].

- **[illegible] ($n = 0$):** [illegible] $|S| = 0$ [illegible] $S = \emptyset \subseteq \mathbb{N}^{<0}$, [illegible] S [illegible].
- **[illegible] ($n \to n + 1$):** [illegible] $P(n)$ [illegible] $S \subseteq \mathbb{N}$ [illegible] $|S| = n + 1$.
  - $S \neq \emptyset$, [illegible] [illegible] [illegible] $s \in S$.
  - [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], $|S \setminus \{s\}| = n$.
  - [illegible] [illegible] [illegible], [illegible] m [illegible] [illegible]-$S \setminus \{s\} \subseteq \mathbb{N}^{<m}$.
  - [illegible] $k = \max(m, s + 1)$. [illegible] $S \setminus \{s\} \subseteq \mathbb{N}^{<m} \subseteq \mathbb{N}^{<k}$, [illegible] $s \in \mathbb{N}^{<s+1} \subseteq \mathbb{N}^{<k}$.
  - [illegible] $S \subseteq \mathbb{N}^{<k}$, [illegible] S [illegible].

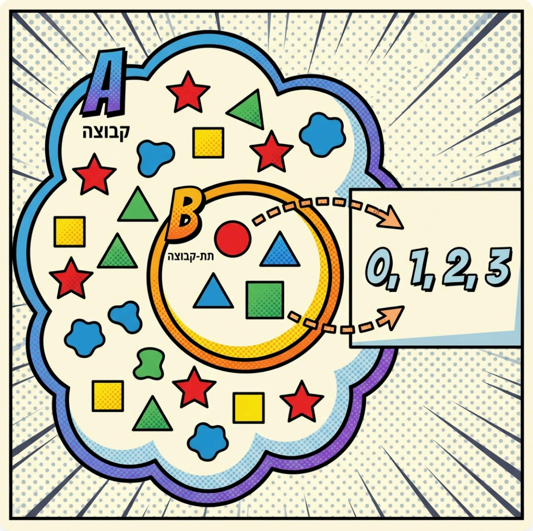
[illegible] [illegible] [illegible] [illegible]: $S \subseteq \mathbb{N}$ [illegible] $\iff$ S [illegible].

[illegible]

[illegible]: [illegible] A [illegible] $B \subseteq A$-[illegible], [illegible] B [illegible].

[illegible]:

- [illegible] $f\colon A \to \mathbb{N}^{<n}$ [illegible]"[illegible] [illegible] ([illegible] A [illegible]).
- [illegible] $f[B] = \{f(b) \mid b \in B\} \subseteq \mathbb{N}^{<n}$.
- $f[B]$ [illegible], [illegible] $\mathbb{N}$-[illegible], $f[B]$ [illegible].
- [illegible] $f|_B\colon B \to f[B]$ [illegible]"[illegible], [illegible] $B \sim f[B]$.
- [illegible], B [illegible].


A
קבוצה
B
תת-קבוצה
0,1,2,3

[illegible] [illegible] ([illegible])

[illegible] [illegible] [illegible] ([illegible]):

[illegible] A [illegible], [illegible] $c \in A$ [illegible], [illegible] $g : \mathbb{N} \times A \to A$ [illegible]. [illegible] [illegible] [illegible] [illegible] $f : \mathbb{N} \to A$ [illegible]:

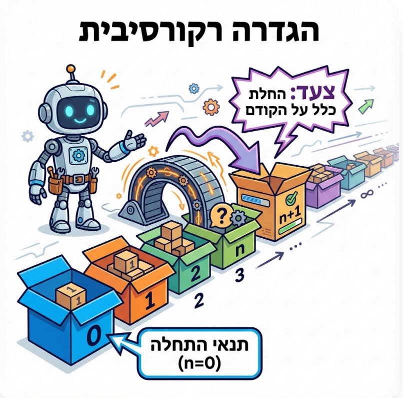
1. [illegible] [illegible]: $f(0) = c$
2. [illegible] [illegible]: [illegible] $n \in \mathbb{N}$, $f(n+1) = g(n, f(n))$.

[illegible] [illegible]:

- [illegible], [illegible] [illegible] $a_0, a_1, a_2, \ldots$ [illegible] [illegible]-A [illegible] [illegible] $a : \mathbb{N} \to A$, [illegible] $a(n) = a_n$.
- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible] ([illegible]):

- [illegible] [illegible] $f(n) = n!$ [illegible]:
  - $f(0) = 1$
  - $f(n+1) = (n+1) \cdot f(n)$


הגדרה רקורסיבית
צעד: החלת כלל על הקודם
n+1
?
n
3
2
1
0
∞
תנאי התחלה (n=0)

[illegible]

[illegible]:

- [illegible] f_1, f_2 [illegible].
- [illegible]-$f_1(n) = f_2(n)$ [illegible] n.
  - [illegible]: $f_1(0) = c = f_2(0)$.
  - [illegible]: [illegible] $f_1(n) = f_2(n)$. [illegible]:

$$f_1(n+1) = g(n, f_1(n)) = g(n, f_2(n)) = f_2(n+1)$$

- [illegible] $f_1 = f_2$.

[illegible] ([illegible]):

- [illegible] "[illegible]" [illegible] k [illegible] $h_k : \{0, \dots, k\} \to A$ [illegible] $n < k$.
- [illegible] k [illegible] h_k.
- [illegible]: $h_0 \subset h_1 \subset h_2 \subset \dots$
- [illegible] $f = \bigcup\{h_k \mid k \in \mathbb{N}\}$.
- f [illegible] n [illegible].

[illegible]: [illegible]’

PHI
1,1,2,3,5,8,13,21
LEONARDO PISANO
(FIBONACCI)

'
(c. 1170 – c. 1250)

[illegible]: [illegible]’[illegible]: $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$.

[illegible]: [illegible] $n \in \mathbb{N}$, [illegible] $F_n < 2^n$.

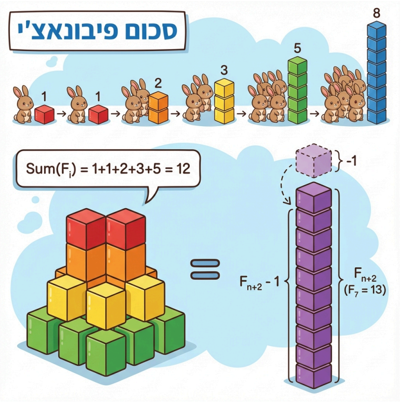
[illegible]:

- [illegible] $A = \{n \in \mathbb{N} \mid F_n < 2^n\}$.
- [illegible] $n \in \mathbb{N}$. [illegible]-$\mathbb{N}^{<n} \subseteq A$.
- [illegible]-$\mathbb{N}^{<n+1} \subseteq A$:
- [illegible]-$\mathbb{N}^{<n+1} = \mathbb{N}^{<n} \cup \{n\}$, [illegible]-$n \in A$:
  - [illegible] $n = 0$: $0 < 1$ [illegible]
  - [illegible] $n = 1$: $1 < 2$ [illegible]
  - [illegible] $n \geq 2$: $F_n = F_{n-1} + F_{n-2}$

$$< 2^{n-1} + 2^{n-2} < 2^{n-1} + 2^{n-1} = 2^n$$

- [illegible]-$n \in A$, [illegible] $\mathbb{N}^{<n+1} \subseteq A$.
- [illegible] $A = \mathbb{N}$.

[illegible]: [illegible] [illegible] [illegible] [illegible]’


סכום פיבונאצ'י
1
1
2
3
5
8
Sum(F$_i$) = 1+1+2+3+5 = 12
-1
F$_{n+2}$ - 1
F$_{n+2}$ (F$_7$ = 13)

[illegible]: [illegible] [illegible]’ [illegible]: $F_0 = 0, F_1 = 1, F_{n+2} = F_{n+1} + F_n$.

[illegible]: $\sum_{i=0}^{n} F_i = F_{n+2} - 1$.

[illegible] [illegible]:

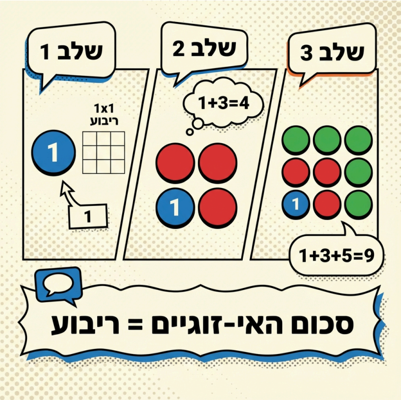
- [illegible] $(n = 0)$: [illegible]: $F_0 = 0$. [illegible]: $F_2 - 1 = 1 - 1 = 0$. [illegible].
- [illegible]: [illegible] [illegible] [illegible] n, [illegible] $\sum_{i=0}^{n} F_i = F_{n+2} - 1$.
- [illegible] [illegible] $n + 1$:

$$\sum_{i=0}^{n+1} F_i = \left(\sum_{i=0}^{n} F_i\right) + F_{n+1} = (F_{n+2} - 1) + F_{n+1}$$

$$= (F_{n+2} + F_{n+1}) - 1 = F_{n+3} - 1 = F_{(n+1)+2} - 1$$

- [illegible] [illegible] [illegible] [illegible] n.

[illegible]: [illegible] [illegible] [illegible]-[illegible]


שלב 1
1x1
ריבוע
1
1
שלב 2
1+3=4
1
שלב 3
1
1+3+5=9
סכום האי-זוגיים = ריבוע

[illegible]: [illegible] $S(n)$ [illegible] n [illegible] [illegible]-[illegible] [illegible] [illegible] [illegible]:

- **[illegible] [illegible]:** $S(0) = 0$
- **[illegible]:** $S(n+1) = S(n) + (2n+1)$

[illegible]: [illegible] $n \geq 0$, $S(n) = n^2$.

[illegible] [illegible]:

- **[illegible]:** $S(0) = 0 = 0^2$. [illegible].
- **[illegible]:** [illegible] $S(n) = n^2$. [illegible] [illegible] $n+1$:

$$S(n+1) = S(n) + (2n+1) = n^2 + 2n + 1 = (n+1)^2$$

- [illegible] [illegible] [illegible] [illegible] n.

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible]

[illegible]: [illegible] [illegible] [illegible] $n \geq 2$ [illegible] [illegible] [illegible] 2 ([illegible]).

[illegible] [illegible] [illegible] [illegible] [illegible]:

[illegible] [illegible] [illegible] $n \geq 2$ [illegible] [illegible] [illegible] [illegible] 2.

- [illegible] $A = \{n \geq 2 : n \text{ 2 [illegible] [illegible] [illegible] [illegible]}\}$.
- [illegible] [illegible], $A \neq \emptyset$.
- [illegible] [illegible] [illegible] [illegible], [illegible]-A [illegible] [illegible] [illegible], [illegible] m.
- [illegible] [illegible]-$m \geq 2$, [illegible] [illegible] $m = 2q + r$ [illegible] $r \in \{0, 1\}$ [illegible]-$1 \leq q < m$.
- [illegible] $q = 1$: $m = (1r)_2$ - [illegible] [illegible] [illegible]-$m \in A$!
- [illegible] $q \geq 2$: [illegible] [illegible]-$q < m$ [illegible]-m [illegible]-A, [illegible] [illegible]-$q \notin A$, [illegible] $q = (b_k b_{k-1} \cdots b_1 b_0)_2$.
- [illegible] $m = 2q + r = (b_k b_{k-1} \cdots b_1 b_0 r)_2$ - [illegible] [illegible] [illegible]-$m \in A$!